

Total no. of Pages: .....

Roll no. ....

Degree: BTECH

Semester: THIRD

**MID-SEMESTER EXAMINATION, September, 2024**

Course Title: Database Management System (DBMS)

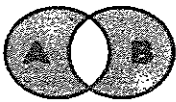
Course Code: COCSC302, CACSC302, CDCSC302, CMCSC302

COCSC05, CACSC05, CDCSC05, CMCSC05 (improvement &amp; backlog) (scaled to 15)

Duration: 1:30 Hours

Max. Marks: 20

**Note:** - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

| Q. No. | Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks | CO |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----|
| 1a     | Explain the difference between DBMS and file system.                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2     | 1  |
| 1b     | Explain the various types of keys used in relational model.                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2     | 1  |
| 2a     | Explain the difference between logical and physical data independence.                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2     | 1  |
| 2b     | What are assertions? Give suitable example for the same.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2     | 2  |
| 3a     | <p>The relational schema is given below<br/> Worker(empid, empname, empmagr, dept, birthdate, hiredate, salary)<br/> Project(projno, projname, projmgr, budget, startdate, duration)<br/> Assign(projno, empid, hoursassign, rating)<br/> Write the relational algebra expression and SQL statements for the given query<br/> Find the projects and names of all the works assigned to projects with startdate 1feb, 2024.</p>                                                                                               | 2     | 2  |
| 3b     | <p>Using relational schema given in question 3a, write the relational algebra expression and sql statements for the given query<br/> Find the employee names and manager names of the workers who are not assigned to any project</p>                                                                                                                                                                                                                                                                                        | 2     | 2  |
| 4a     | <p>(i) Write the syntax of GROUP BY with an aggregate function.<br/> (ii) Write SQL query to determine how many row(s) a table has</p>                                                                                                                                                                                                                                                                                                                                                                                       | 2     | 2  |
| 4b     | <p>Given two relations A and B. Write the SQL query(s) that depict the following join as given in Fig A and Fig B. Explain through examples.</p> <div style="display: flex; justify-content: space-around; align-items: center;"> <div style="text-align: center;">  <p>Fig A</p> </div> <div style="text-align: center;">  <p>Fig B</p> </div> </div> | 2     | 2  |
| 5a     | Define two integrity constraints with suitable examples.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2     | 2  |
| 5b     | <p>Let R and S be two relations with 'm' and 'n' number of tuples respectively. What will be minimum and maximum number of tuples in</p> <ol style="list-style-type: none"> <li>Natural join of R and S</li> <li>R X S</li> </ol>                                                                                                                                                                                                                                                                                            | 2     | 2  |